

Co-author update: May-September 2026 revision and the human-impact proxy decision

Internal discussion document for Suzette, Vivian, and Ondrej

2026-09-27

Purpose and status. This memo brings Suzette and Vivian up to date on the scientific revision completed since May 2026. Sections 1–11 describe canonical R1 and its registered validation evidence. Section 12 introduces a later exploratory analysis using KK10 and HYDE that is not yet part of the manuscript or reviewer response. The manuscript itself is unchanged by this memo.

1 Executive summary

The revision since May is a substantial analytical reconstruction rather than a cosmetic rewrite. The current R1 package now exposes the response trajectories, distinguishes analysis-specific subsets, corrects the extraction and interpretation of HVarPart components, controls time and spatial structure reciprocally, and reports coordinated sensitivity checks.

Under the **canonical R1 specification**, square-root SPD is the primary human predictor and four palaeoclimate variables form the climate block. After controlling time within sequences, climate has the larger zero-truncated contribution in 891 of 1,098 spatial sequence models (81.1%). After controlling spatial structure, climate is larger in 69 of 70 estimable continental-region-age comparisons. H2 similarly attributes more of the broad change in PAP interrelationships to climate than to SPD in most estimable groups.

The revision also narrows what this result means. The HVarPart allocations divide variation attributed to predictors within fitted models; they are not percentages of all ecological change. The result concerns pollen assemblage properties, broad analytical regions, and 8.5–2.0 ka BP. It does not establish that people were absent, locally unimportant, or ecologically benign.

The registered validation analyses generally retain the canonical aggregate ordering, but identify important qualifications: model fit varies widely, several PAPs are redundant, residual dependence remains in some subsets, and individual results can change with spatial thinning, SPD radius, or proxy definition.

A later exploratory analysis adds KK10 land use and HYDE population to the human block after local collinearity filtering. That evidence materially challenges the robustness of the paper-wide climate-dominance headline. Because Suzette and Vivian have not yet seen the canonical revision, the exploratory result and the manuscript decision are deliberately deferred to the final section.

Table 1: Revision map from May to September 2026.

Period	Change	Why it matters
May-June	Pollen-count extraction and retention diagnostics	Separated data coverage from model robustness and documented the precision-coverage trade-off.
June-July	Ten PAP trajectories refitted and exported	Made response histories visible before attribution and prevented long sequences from dominating summaries.
June-July	PAP correlation and reduced-response analysis	Tested whether overlapping assemblage properties determined H1.
July-August	HVarPart extraction corrected and model fit separated	Replaced the old ratio with commonality, signed components, display compositions, and adjusted R-squared.
August	Time and spatial controls added to H1	Made the spatial and temporal comparisons reciprocal and explicit about dependence.
August-September	Dependence, thinning, omission, radius, and event checks	Tested sampling structure and alternative definitions on matched subsets.
September	R1 manuscript and evidence registry rebuilt	Aligned scope, terminology, figures, replies, and reproducible public evidence.
Late September	KK10 and HYDE added as exploratory human proxies	Exposed sensitivity to the representation of human impact; this is not yet in R1.

2 What changed since May 2026

The table below is a scientific map of the revision, not a list of software commits. Each step either answered a reviewer concern, corrected an analytical interpretation, or made the evidential chain reproducible.

Three evidence states must remain separate throughout this memo:

1. **Canonical R1:** analyses and figures already incorporated in the revised manuscript and response.
2. **Registered validation:** supplementary checks that qualify, but do not silently redefine, the canonical specification.
3. **Exploratory proxy decision:** the later KK10/HYDE comparison, not yet incorporated in R1.

The project also moved from loosely connected scripts to registered, restartable pipelines with public outputs, analysis profiles, source hashes, and a claim-evidence registry. That workflow change does not itself alter a scientific conclusion, but it now makes every value and figure in R1 traceable to a stable source.

Table 2: Analysis-specific populations used by the revised evidence.

Analysis	Unit	Number
Preparation pool	pollen sequences	1270
Initial spatial H1 inventory	pollen sequences	1262
Estimable initial spatial H1	pollen sequences	1156
Controlled spatial H1	pollen sequences	1098
Controlled temporal H1	continental-region-age comparisons	70
Matched SPD-radius spatial comparison	pollen sequences	999
Matched expert-event spatial comparison	pollen sequences	151

3 Data scope, analytical populations, and the study interval

The preparation pool contains 1,270 pollen sequences. The initial spatial inventory contains 1,262, but eligibility differs among analyses because responses, predictors, temporal coverage, controls, and rank requirements differ. The revised manuscript therefore no longer implies that every sequence contributes to every result.

The principal SPD comparison is restricted to 8.5–2.0 ka BP. The younger boundary is a practical common boundary arising from archaeological coverage, not a universal statement that SPD becomes invalid after 2 ka BP. Expert-event histories extend later in some regions, but are interpreted separately.

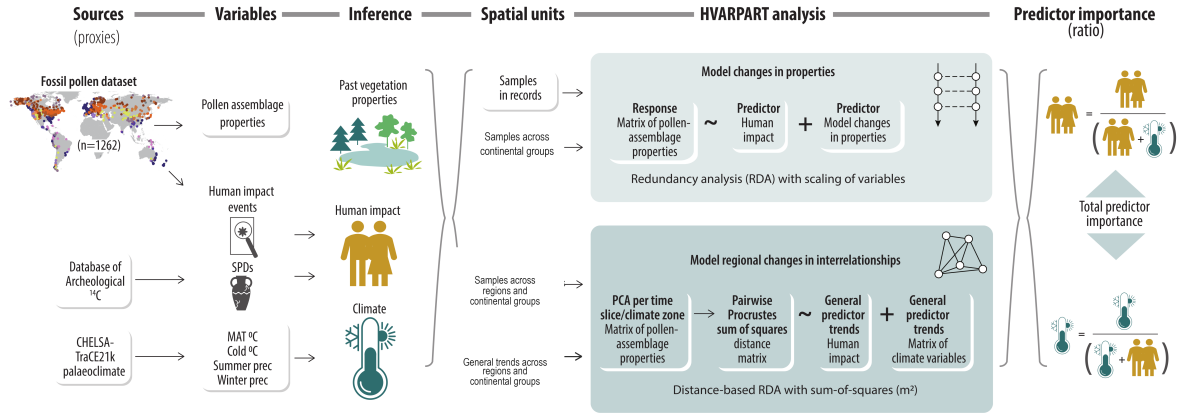


Figure 1: Current R1 study overview. The figure distinguishes the data sources, analytical scales, and the two hypotheses.

4 Pollen counts and coverage

The baseline profile retained 1,270 sequences. Increasing the nominal count target leaves sequences at 300 grains, at 500 grains, and at 1,000 grains. Losses are uneven among continental regions.

This analysis supports retaining the baseline rule while acknowledging a precision-coverage trade-off. It is a **coverage diagnostic**, not a rerun of H1, and therefore cannot establish that H1 estimates are invariant to stricter pollen thresholds.

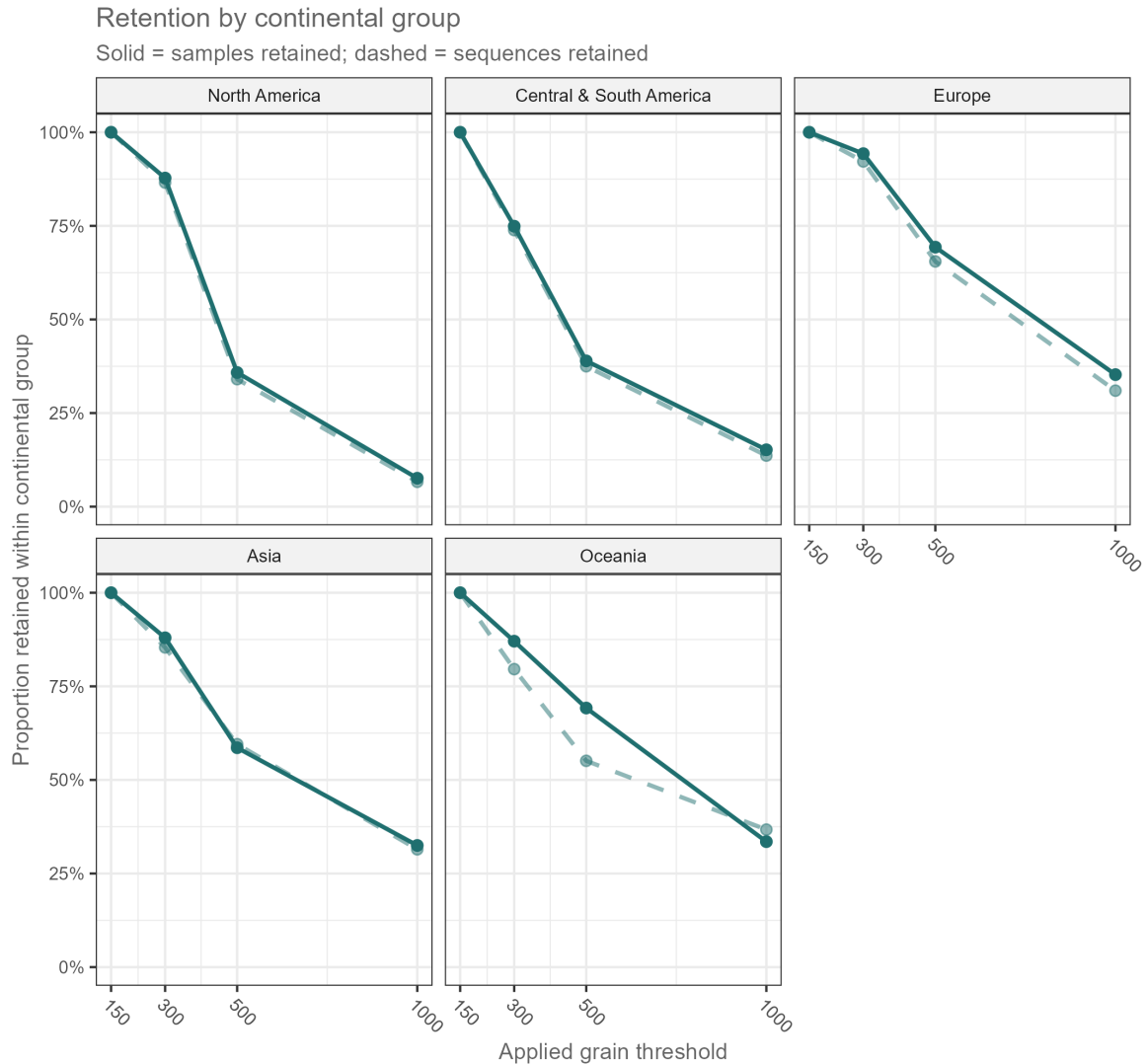


Figure 2: Pollen-count diagnostics used in R1. Retention profiles show the increasingly severe and geographically uneven data loss under higher count targets.

5 Temporal patterns in the ten pollen assemblage properties

The revised response framework contains ten PAPs rather than nine, including the difference between Shannon- and Simpson-weighted diversity. All temporal hGAMs were refitted through a reproducible request-and-export workflow. The displayed trajectories are equal-weighted summaries of sequence-level posterior predictions within observed temporal coverage, so long or densely sampled sequences do not dominate the regional curves.

The main message is heterogeneity rather than one global temporal trajectory. Richness, diversity, evenness, turnover, rates of change, and change-point densities do not move synchronously in every region. Missing segments indicate inadequate observed coverage, not an absence of ecological change.

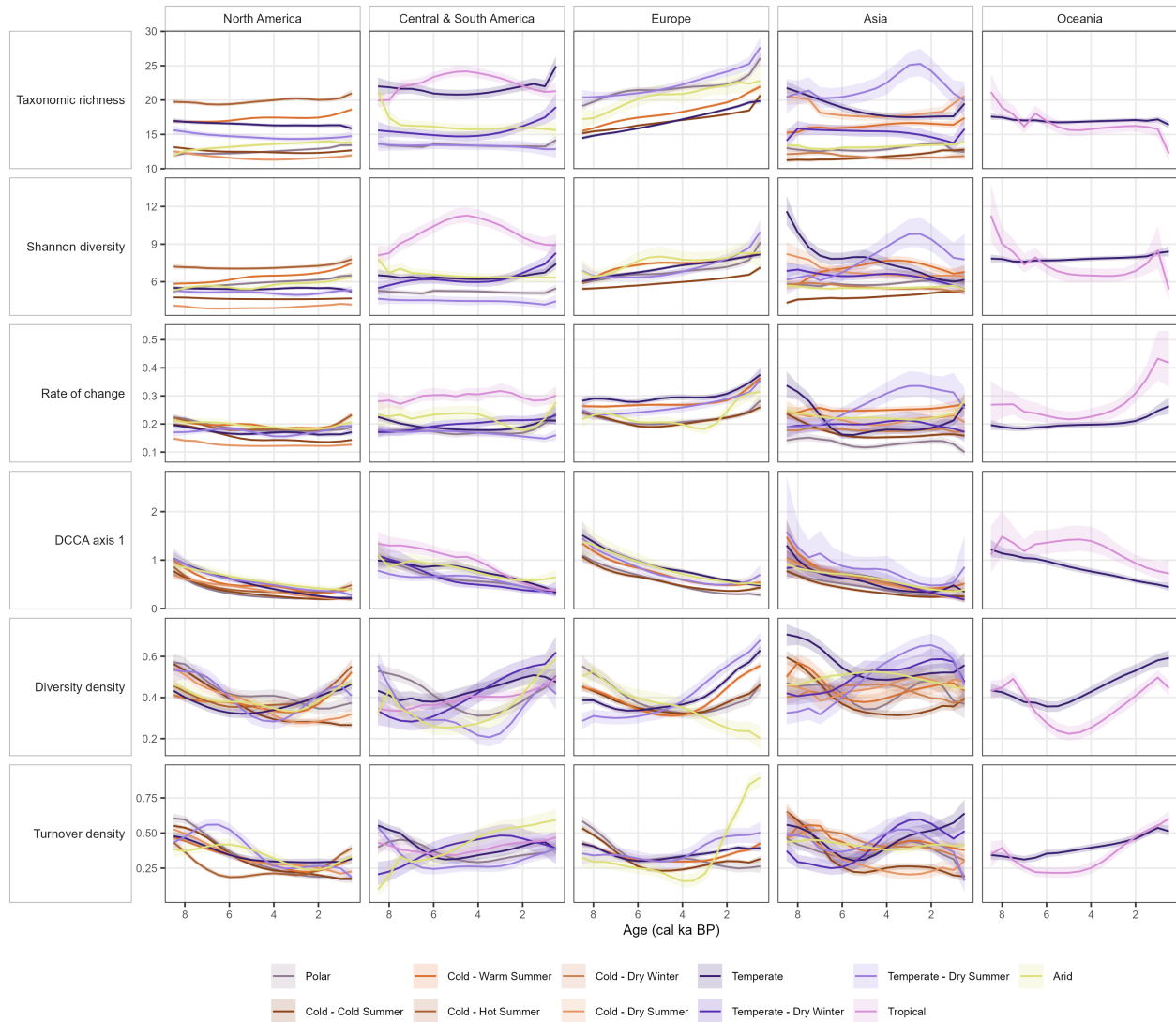


Figure 3: Canonical R1 Figure 2: selected temporal trajectories of pollen assemblage properties. The Supplementary Information shows all ten PAPs.

6 PAP correlation and the reduced response

Correlation among PAPs is expected because several properties share taxa, abundance information, or mathematical components. Of 45 PAP pairs, 3 have a median absolute correlation of at least 0.8 across eligible groups; the maximum median absolute correlation is 0.96.

The reduced-response sensitivity removes redundant PAPs and compares only sequences estimable under both configurations. Across 1,114 matched sequences, the aggregate climate-over-human ordering persists, although individual allocations change. This supports the broad canonical ordering; it does not make the ten PAPs independent replicates or imply identical estimates for every sequence.

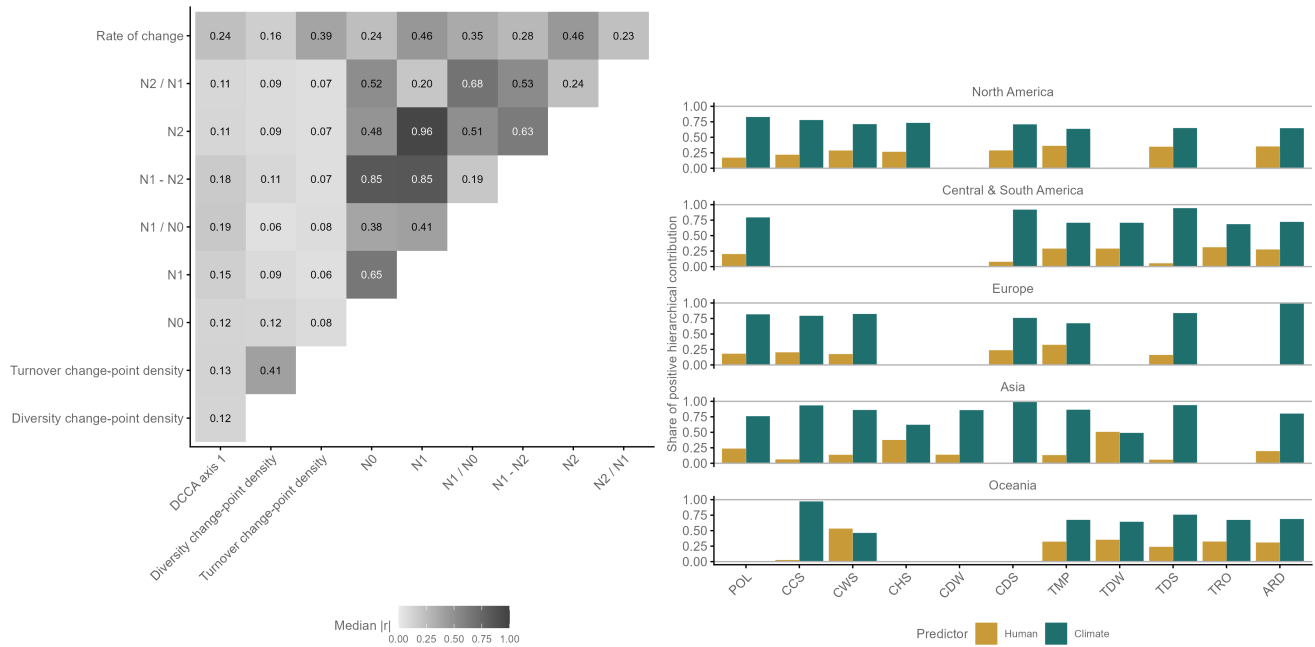


Figure 4: PAP correlation structure (left) and matched full- versus reduced-response H1 allocations (right).

7 What was corrected in HVarPart

The original extraction treated relative importance too simply. The revised workflow starts from the complete commonality decomposition and keeps four concepts separate:

- **Total adjusted (R^2):** how much multivariate response variation the full model explains.
- **Unique adjusted (R^2):** non-shared variation associated with one predictor group after the others are controlled.
- **Signed hierarchical contributions:** complete contributions that retain negative components arising from suppression or collinearity.
- **Zero-truncated composition:** a display transformation that sets negative components to zero and recomposes the remaining contributions to sum to one.

The zero-truncated bars in the main figures are therefore not percentages of all ecological change and are not estimates of pure unique variation. An additional profile excludes models containing negative individual components. The numerical allocations differ among profiles, but the canonical aggregate ordering is retained.

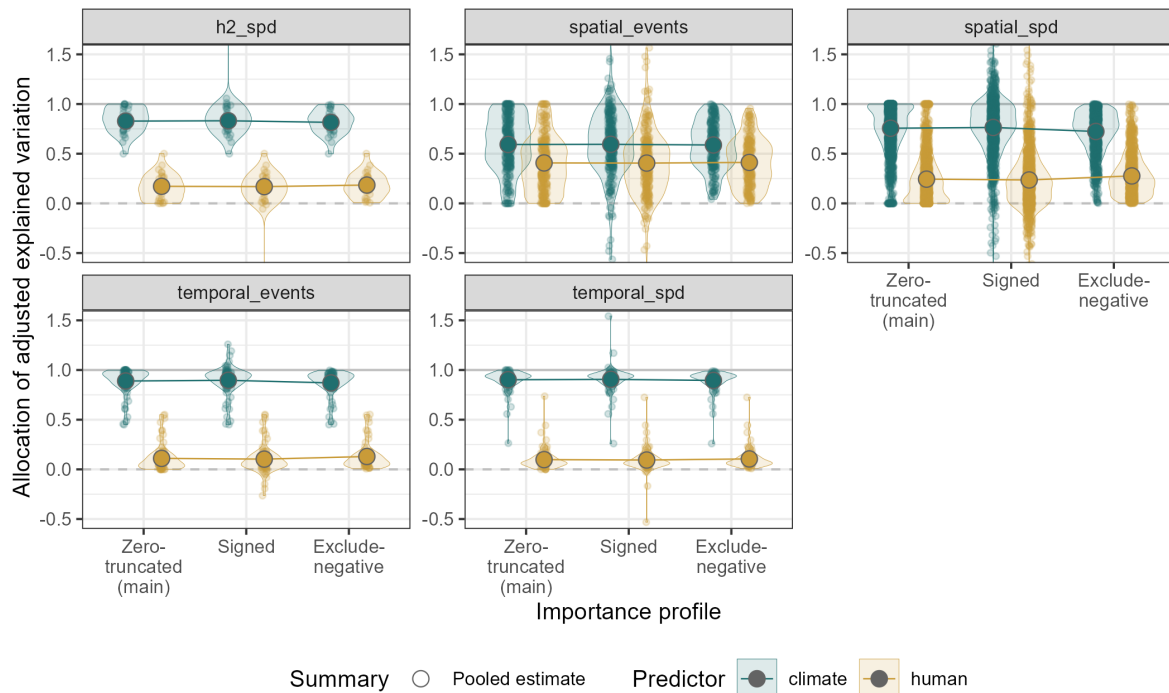


Figure 5: Registered comparison of signed, zero-truncated, and exclude-negative HVarPart profiles. These are alternative summaries of the same decompositions, not interchangeable effect estimates.

8 Model fit is separate from predictor allocation

Of the 1,262 sequences considered for the initial spatial analysis, 1,156 have an estimable decomposition. Their mean total adjusted (R^2) is 0.352 [0.326, 0.374] (mean and region-block bootstrap 95% confidence interval).

The controlled spatial subset contains 1,098 sequences. Its mean adjusted (R^2) is 0.440 [0.426, 0.449], while its median and interquartile range are 0.450 [0.326, 0.567].

Fit varies substantially among sequences and is only weakly related to signed human allocation. A model can allocate most of its explained component to one predictor group while explaining little response variation overall; the reverse is also possible.

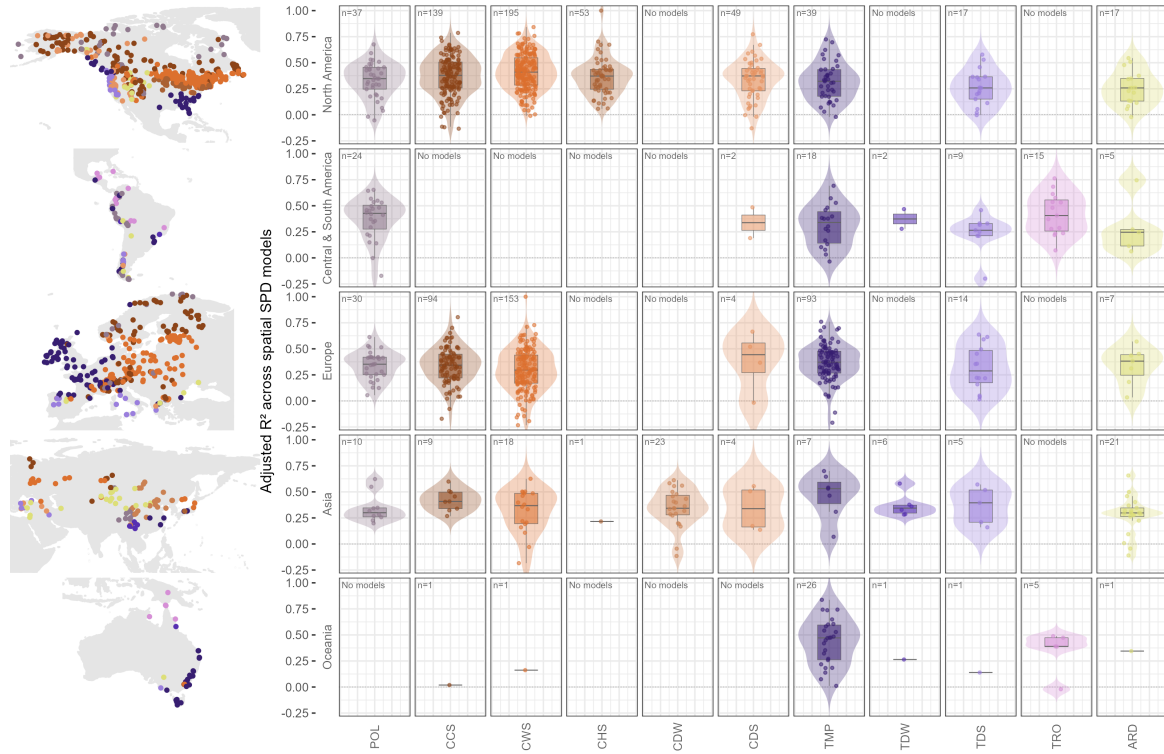


Figure 6: Distribution of total adjusted R-squared across spatial sequence models. Model fit is reported separately from the allocation of explained variation.

9 Canonical H1 spatial result: time controlled within sequences

The revised spatial H1 is fitted within each pollen sequence with time included as a protected control group. Human and climate contributions are then aggregated spatially with the existing dbMEM procedure. This prevents temporal structure within a sequence from being silently attributed to either focal predictor group.

Climate has the larger zero-truncated contribution in 891 of 1,098 eligible sequences (81.1%). The result is heterogeneous: some sequences and regions show stronger human allocations, but the canonical distribution is climate-dominant overall.

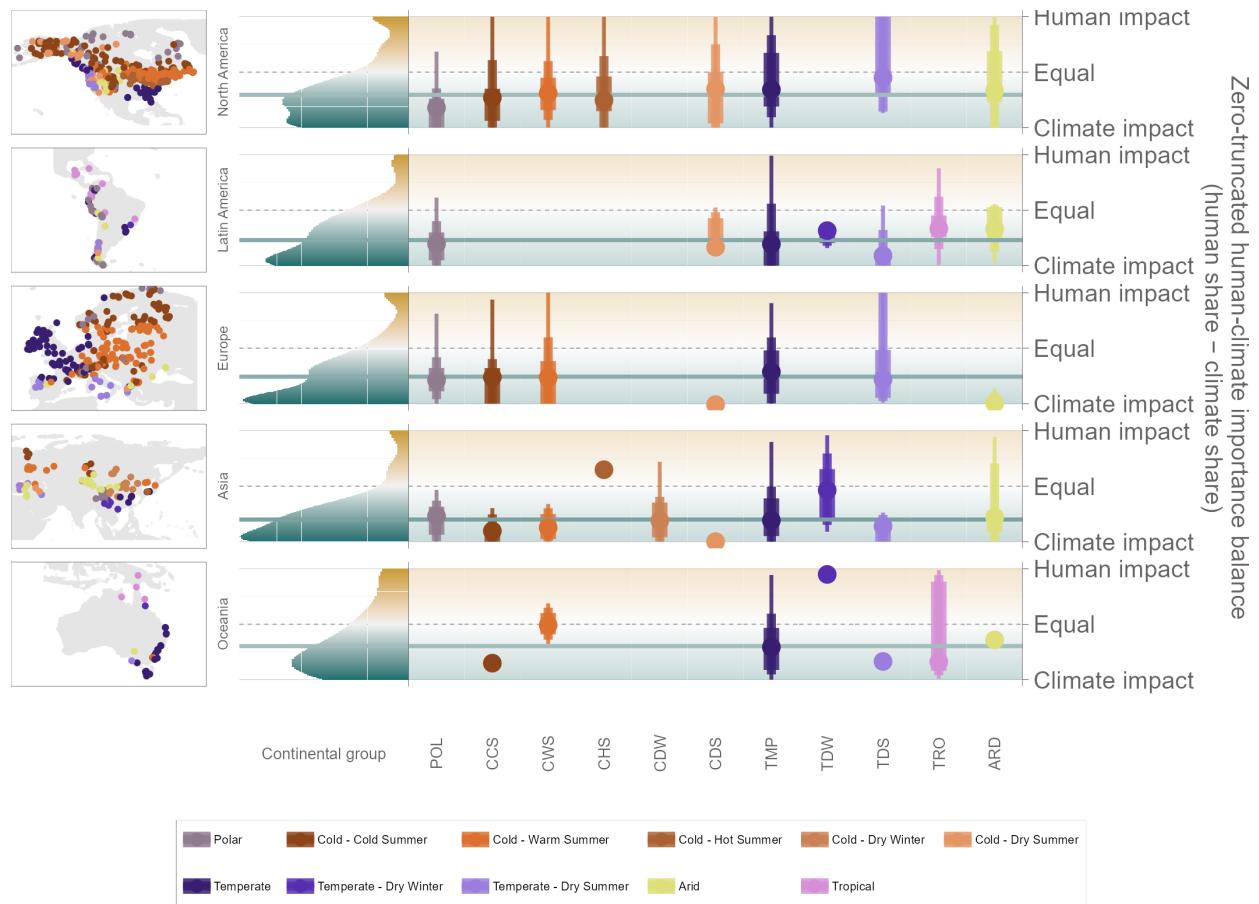


Figure 7: Canonical R1 Figure 3. Sequence-level human-minus-climate balance after controlling time within sequences and aggregating spatial pattern. Gold denotes greater human allocation and teal denotes greater climate allocation.

10 Canonical H1 temporal result: spatial structure controlled within time slices

The revised temporal H1 groups sequences within 500-year continental-region time slices. Selected dbMEM variables represent spatial structure among the eligible sequences before contributions from SPD and climate are interpreted. Non-estimable combinations remain missing rather than being assigned zero effect.

Climate has the larger zero-truncated contribution in 69 of 70 estimable continental-region-age comparisons. Signed and unique-fraction counterparts identify time slices where negative components, shared variation, or weak unique signals make the display composition incomplete.

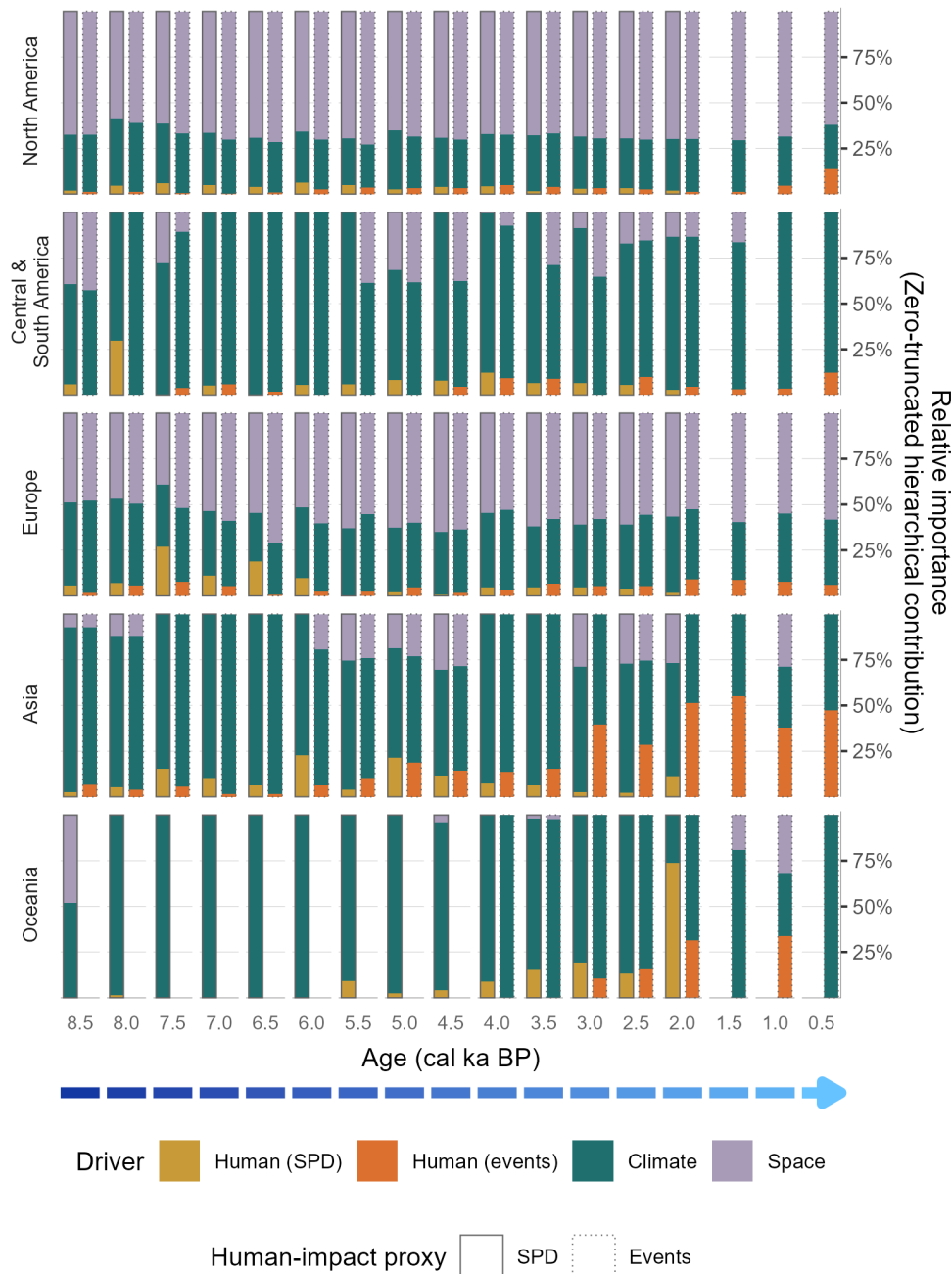


Figure 8: Canonical R1 Figure 4. Human, climate, and spatial allocations through time after spatial control within each continental-region time slice. Missing combinations are non-estimable.

11 Canonical H2: changing interrelationships among PAPs

H2 asks whether human activity is the predominant predictor of changes in relationships among PAPs. PCA and symmetric Procrustes distances summarise changes in collective PAP covariance between time slices within regions. Regional SPD and climate trajectories then enter a second HVarPart analysis.

The canonical analysis attributes more of the broad change in PAP interrelationships to climate than to SPD in most estimable groups. The zero-truncated analysis-level climate allocation is 0.828.

This ordination is collective and multivariate. It cannot identify which pollen taxon changed, and missing or short trajectories can reflect incomplete estimability rather than ecological stability. H2 remains based on SPD alone; it has not been harmonised with KK10 and HYDE.

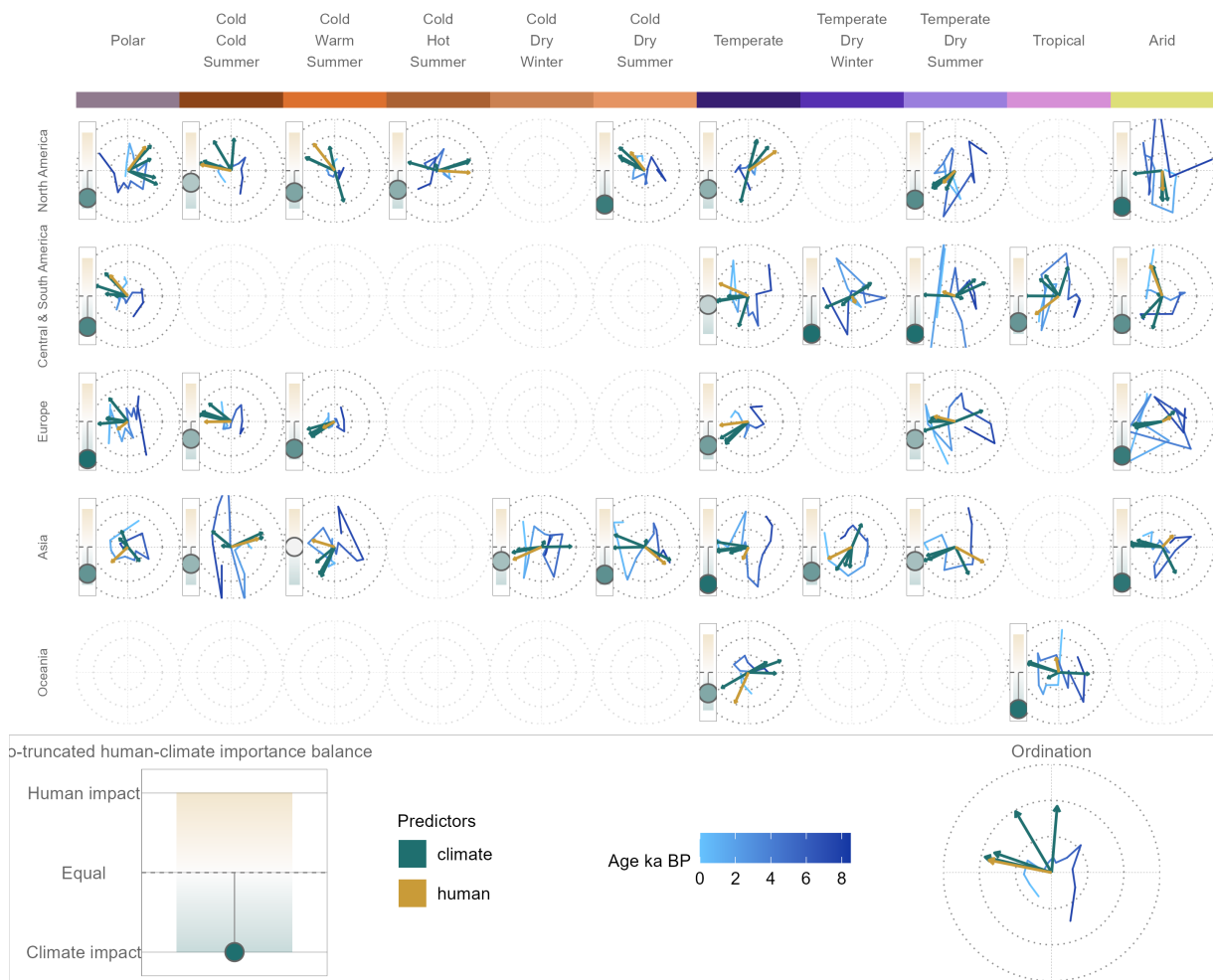


Figure 9: Canonical R1 H2 figure. Ordination trajectories describe changes in collective relationships among PAPs; predictor arrows are not taxon-level effects.

12 Dependence, thinning, and omission checks

Residual Moran diagnostics find positive spatial autocorrelation in 9 of 14 unfiltered comparisons and 4 of 14 residual comparisons after spatial control. Observed Moran's (I) ranges from -0.068 to 0.131 before control and from -0.061 to 0.054 afterwards.

Spatial thinning retains a median of 361 sequence models at 250 km and 216 at 500 km across 100 replicates. Leave-one-continental-region and leave-one-region comparisons generally retain a climate-dominant aggregate balance, while showing which omissions increase uncertainty. These diagnostics reduce the risk that dense sampling or one geographical group determines the result; they do not remove all dependence or establish causality.

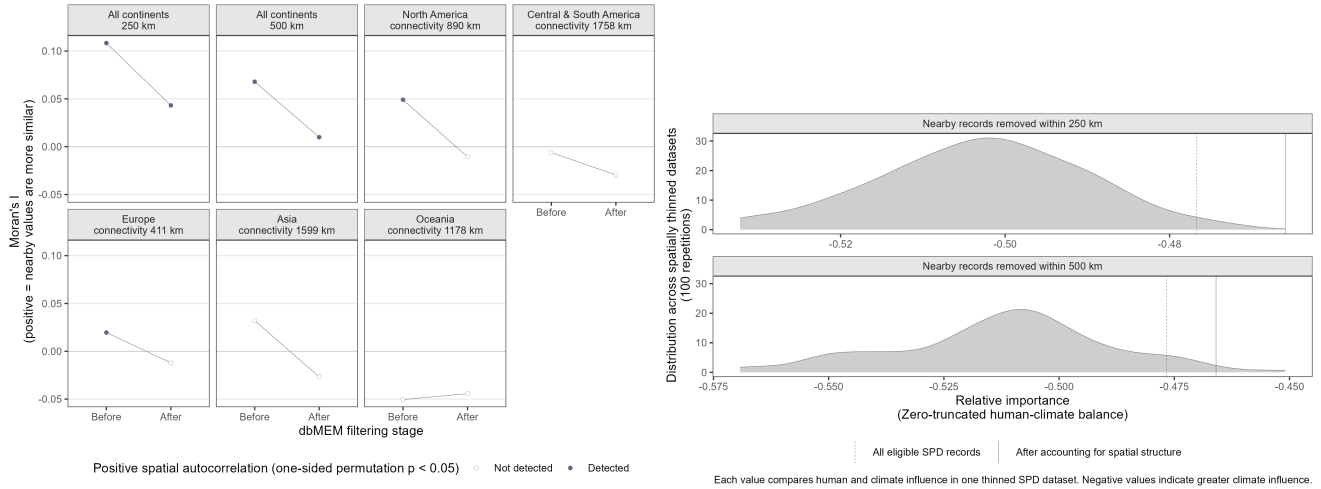


Figure 10: Residual Moran diagnostics before and after spatial control (left) and spatial-thinning sensitivity (right). Leave-one-group-out results are summarised in the text.

13 SPD-radius sensitivity

The primary SPD workflow uses archaeological dates within 250 km and a 500 km fallback where the strict neighbourhood has insufficient dates. The matched sensitivity compares strict 250 km and 500 km predictors on identical analytical units.

Among 999 matched spatial sequences, 963 have defined balances under both radii. The median 500-km-minus-250-km balance change is 0.025, and 94 sequences reverse their human-versus-climate ranking. At the temporal scale, 3 of 70 matched comparisons reverse ranking.

The fallback improves coverage and does not overturn the aggregate canonical conclusion, but it changes some individual estimates. Neither radius is equivalent to the pollen-source area, so this is a neighbourhood-choice sensitivity rather than a scale-matched archaeological comparison.

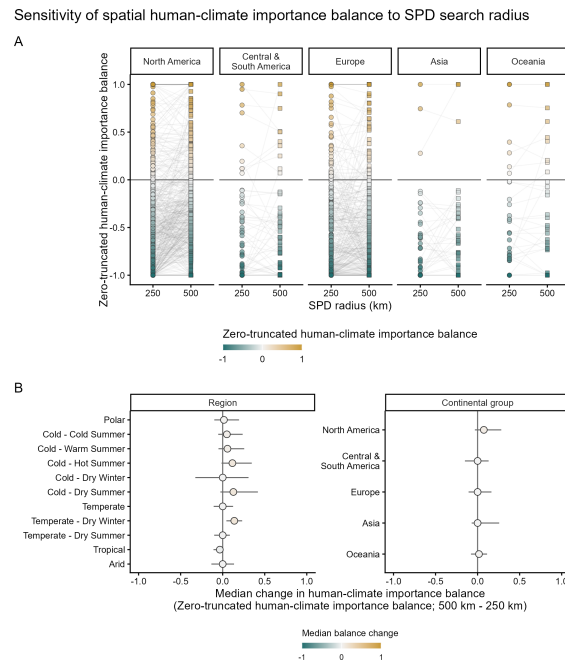


Figure 11: Matched SPD-radius sensitivity across spatial sequence models.

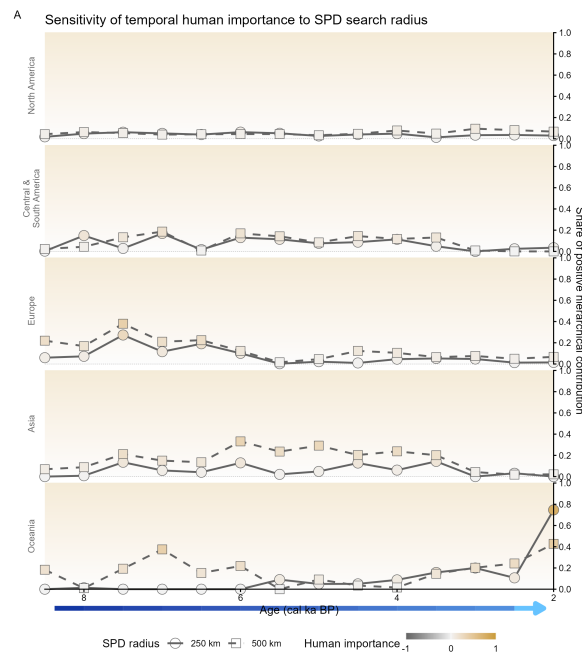


Figure 12: Matched SPD-radius sensitivity across continental-region time slices.

14 Expert-event proxy sensitivity

Expert-screened pollen events contain locally informed evidence that SPD may miss, but they are coarser, geographically uneven, incomplete, and partly circular because event definition and PAP responses can derive from the same pollen sequence. The sensitivity therefore uses matched observed-event subsets rather than default-filled histories.

The matched analysis contains 151 spatial sequences and 61 continental-region-age comparisons. Median spatial human-minus-climate balance is -0.018 for the combined SPD-plus-event model and -0.254 for the event-only model. Temporally, adding events changes median human allocation by 0.034, while substituting events changes it by -0.012.

Variation in both directions shows that human attribution depends partly on proxy definition. The expert chronologies are therefore complementary sensitivity evidence, not independent validation of SPD.

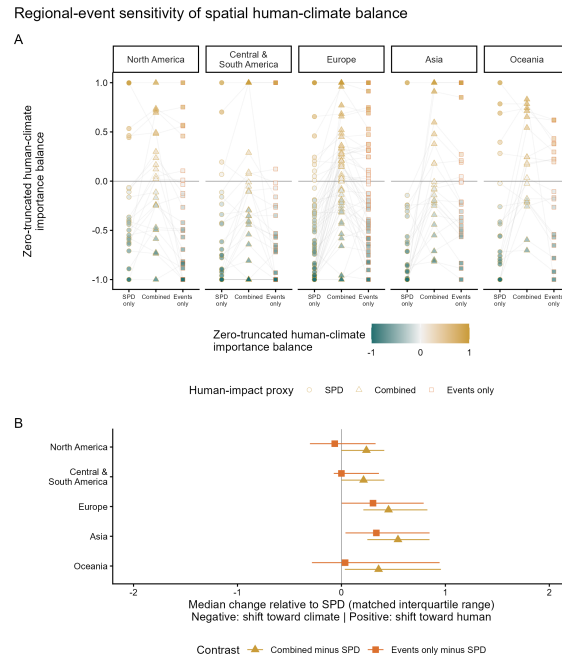


Figure 13: Matched expert-event sensitivity across spatial sequence models.

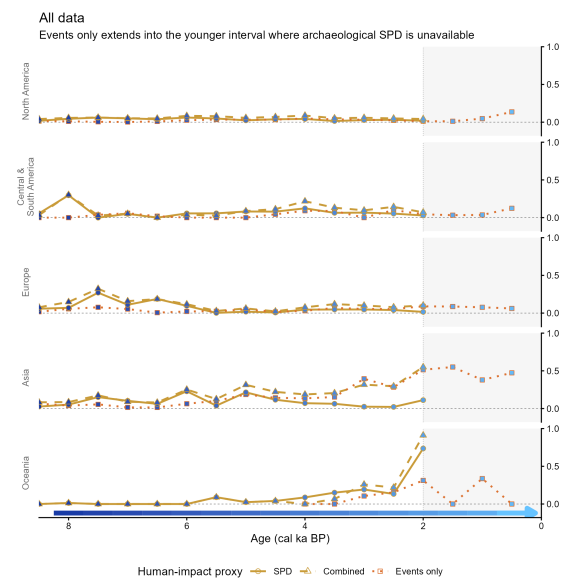


Figure 14: Matched expert-event sensitivity across continental-region time slices.

Table 3: How the submitted narrative changed in canonical R1.

Element	Submitted framing	Current R1 framing
Scope	Holocene vegetation dynamics	Pollen assemblage dynamics from 8.5 to 2.0 ka BP.
H1	Human versus climate importance without reciprocal controls	Time-controlled spatial and spatially controlled temporal comparisons.
Importance	Relative importance presented as a simple ratio	Fit, unique fractions, signed commonality, and display composition kept distinct.
H2	Broad ecological interpretation of changing relationships	Collective PAP covariance with missingness and taxon-level limits explicit.
Human evidence	SPD and event evidence treated more comparably	SPD is primary; expert events are a complementary, partly circular sensitivity.
Conclusion	Broad claim about pre-industrial human importance	Scale- and interval-bounded statement that retains local human histories.

Table 4: Canonical finding and the qualification contributed by each validation analysis.

Check	Canonical finding	Qualification
Pollen-count coverage	Baseline retains the broadest usable sequence set.	Higher count targets sharply and unevenly reduce coverage; H1 was not refitted.
Reduced PAP response	Aggregate climate-over-human ordering persists.	Individual allocations change; PAPs are not independent replicates.
Extraction profiles	Ordering persists across all three summaries.	Signed, zero-truncated, and exclude-negative profiles answer different questions.
Spatial dependence	Controls reduce residual spatial structure.	Some residual dependence remains and limits causal interpretation.
Thinning and leave-out	Aggregate balance is generally retained.	Fewer sequences and particular omissions increase uncertainty.
SPD radius	Aggregate result is less sensitive than individual estimates.	Some sequences and time slices reverse ranking.
Expert events	SPD remains the canonical primary analysis.	Attribution varies with proxy definition; events are incomplete and partly circular.

15 Consequences for the manuscript narrative

The canonical revision supports a more precise paper, but not the broadest wording used in the submitted version. The principal changes are analytical scope and inferential discipline rather than a claim that local human transformation did not occur.

The title now names pollen assemblage dynamics and the 8.5–2.0 ka BP interval. The abstract and Results use analysis-specific denominators rather than presenting the whole collection as one model population. The Discussion distinguishes broad-scale predominance from local human histories, removes unsupported tipping-point and uniformly destructive-impact claims, and states limits arising from pollen counts, spatial mismatch, climate simulations, and proxy construction.

The reviewer response now has a defensible chain from concern to analysis to evidence. However, its current headline still depends on SPD being an adequate primary representation of the human construct. The final section addresses that unresolved point.

16 Final decision: should human impact include SPD, KK10, and HYDE?

16.1 What the additional proxies represent

SPD primarily represents archaeological activity and demography inferred from radiocarbon-date densities. KK10 represents land-use fraction. HYDE represents modelled population. Together they represent a broader human-impact construct than SPD alone, but they remain proxies with different assumptions, resolutions, and uncertainties.

Across the matched 2–8 ka comparison, the convergence dataset contains 15,353 observations from 1,181 pollen sequences and 13 ages. Rank agreement is positive overall: Kendall's tau is 0.87 for SPD versus KK10 and 0.69 for SPD versus HYDE. Regional agreement is less uniform, which is one reason the model performs selection locally.

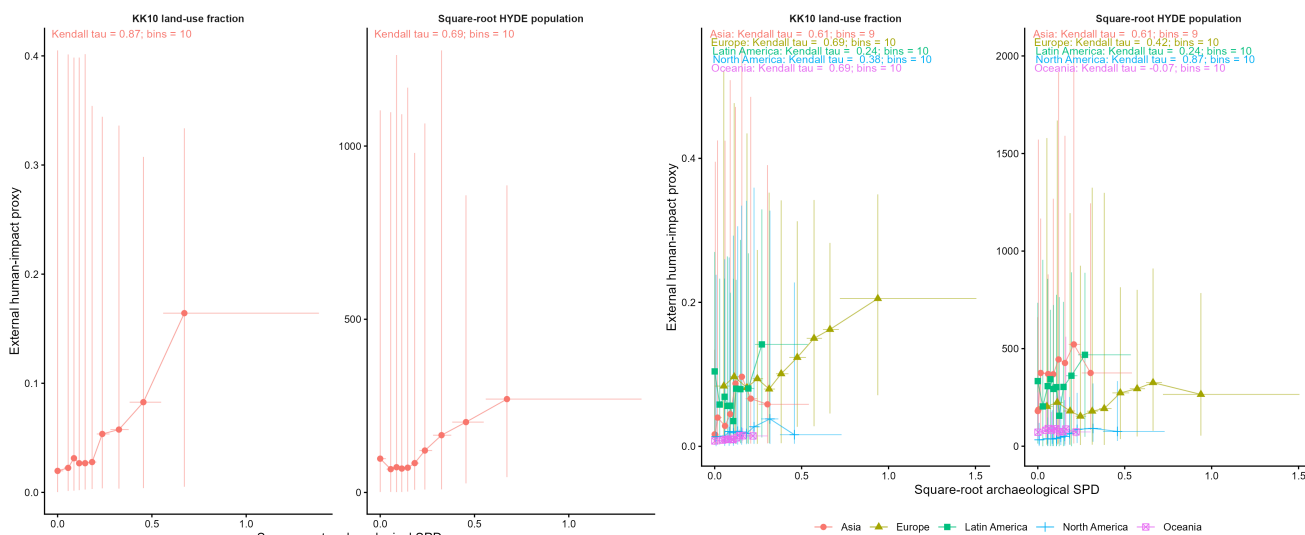


Figure 15: Agreement between SPD and the additional human proxies overall (left) and by continental region (right).

The separate exported spatial atlas shows SPD, KK10, and HYDE for the world and Europe at all thirteen matched time slices. It should be consulted when discussing differences in spatial support, but is not reproduced here because the convergence summaries already establish the evidence needed for this decision.

16.2 Harmonised model design

The exploratory model uses square-root SPD, KK10 fraction, and square-root HYDE as candidate human predictors. The square-root transformations reduce extreme right-skew while retaining ordering; KK10 remains on its natural fractional scale.

Within each sequence or continental-region-age unit, redundant predictors are removed separately within the human and climate blocks using $|r| = 0.8$ and $VIF = 5$. The human preference order is square-root SPD, KK10, then square-root HYDE; the climate order remains annual temperature, coldest-month temperature, warm-season precipitation, then cold-season precipitation. Time and dbMEM controls are protected and added after focal selection. Cross-block correlation is diagnosed but not used for automatic removal because shared human-climate variation is part of the partitioning question.

The retained set can therefore differ among analytical units. Selection reduces within-block redundancy, but it does not prove that a retained variable is causal or that the resulting multivariate human block is uniquely correct.

Three specifications clarify what changes:

1. **Canonical SPD:** square-root SPD, all four canonical climate predictors, 8.5–2.0 ka BP.
2. **Matched SPD bridge:** square-root SPD, the joint model's locally selected climate set, and the same matched 8–2 ka observations as the joint model.
3. **Filtered joint human block:** square-root SPD, KK10, and square-root HYDE after local human filtering, with the same climate selection and cohort as the bridge.

Canonical-to-joint differences combine interval, matched observations, climate selection, and human representation. Bridge-to-joint is the cleaner contrast for the effect of expanding the human block.

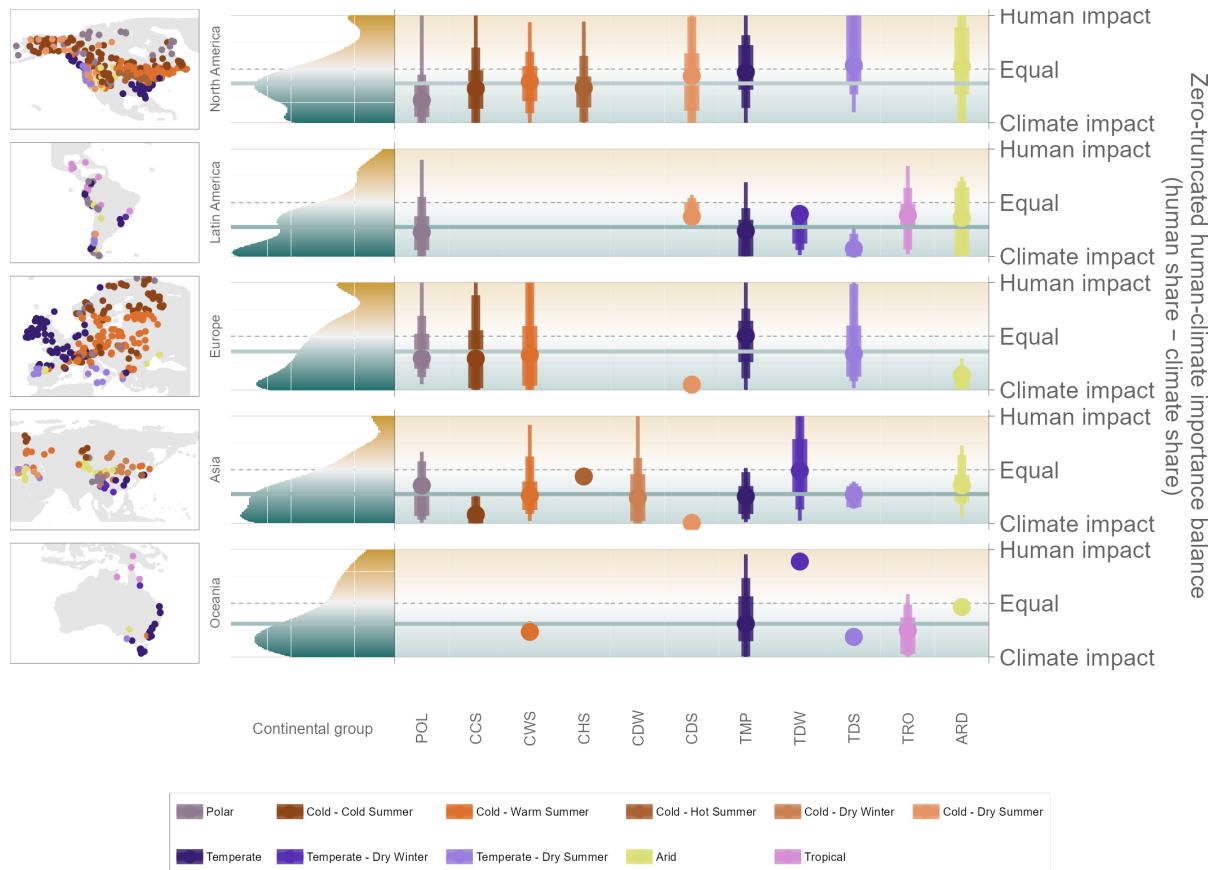


Figure 16: Figure 3 analogue for the matched SPD bridge. Climate remains dominant on the harmonised observations and locally selected climate sets.

16.3 Spatial H1 reverses under the filtered joint block

The exact three-model common spatial subset contains 1,069 sequences. The canonical model is climate-dominant in 869 sequences, and the matched SPD bridge remains climate-dominant in 764.

The filtered joint model is instead human-dominant in 692 sequences and climate-dominant in 376, with 1 tie. The median zero-truncated human-minus-climate balance moves from -0.35 in the bridge to 0.18 in the joint model. Moving from bridge to joint changes 387 sequences from climate- to human-dominant, with no corresponding human-to-climate reversal.

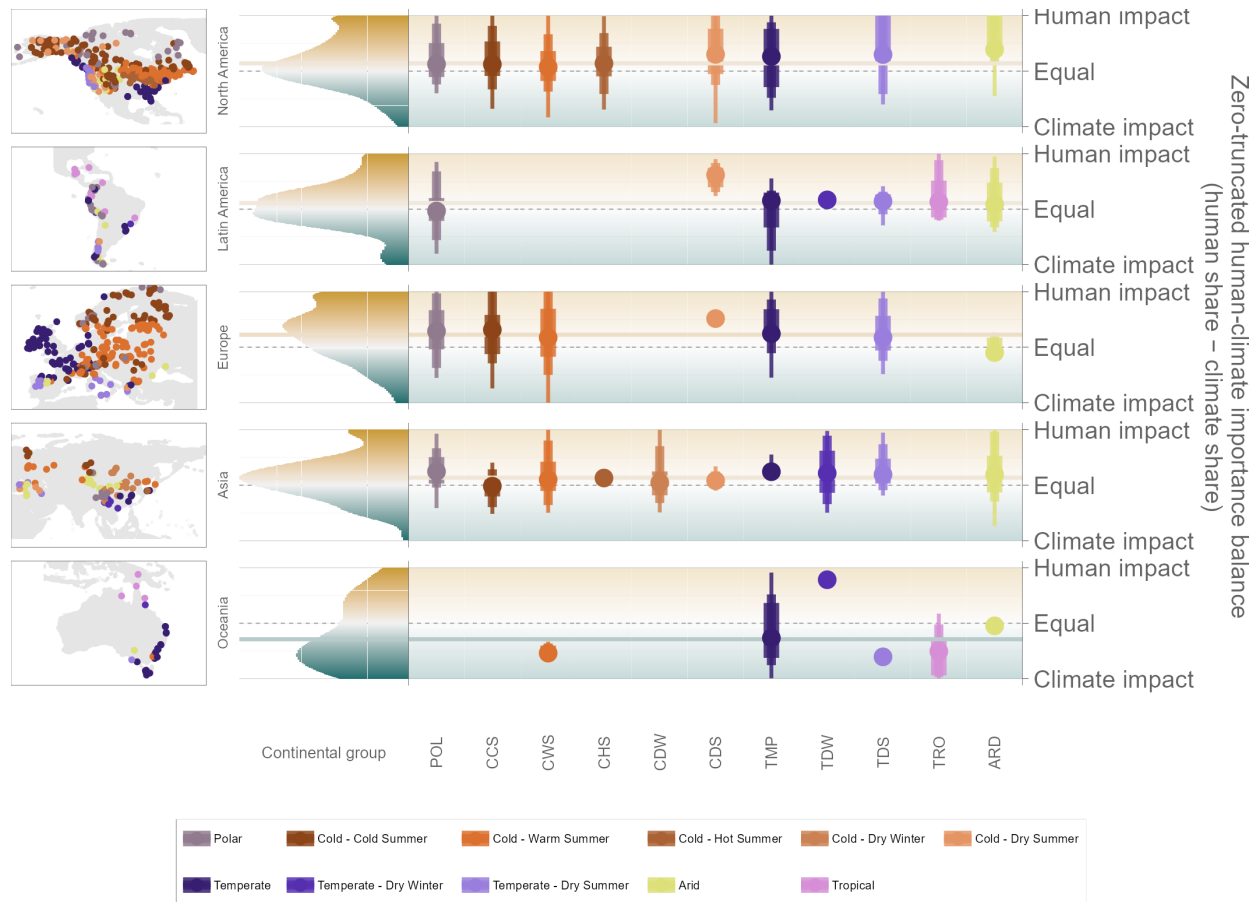


Figure 17: Figure 3 analogue for the filtered joint human block on the exact common subset. This is exploratory evidence and is not the current manuscript Figure 3.

16.4 Temporal H1 weakens but does not fully reverse

All three temporal specifications are compared across the same 65 continental-region-age units. The canonical model is climate-dominant in 64 units and the matched SPD bridge in 65. Under the filtered joint model, climate remains larger in 47 units and humans are larger in 18.

Median human allocation increases from 6% in the bridge to 26% in the joint model. The bridge-to-joint contrast produces 18 climate-to-human reversals. Thus the temporal conclusion weakens substantially but remains climate-dominant overall.

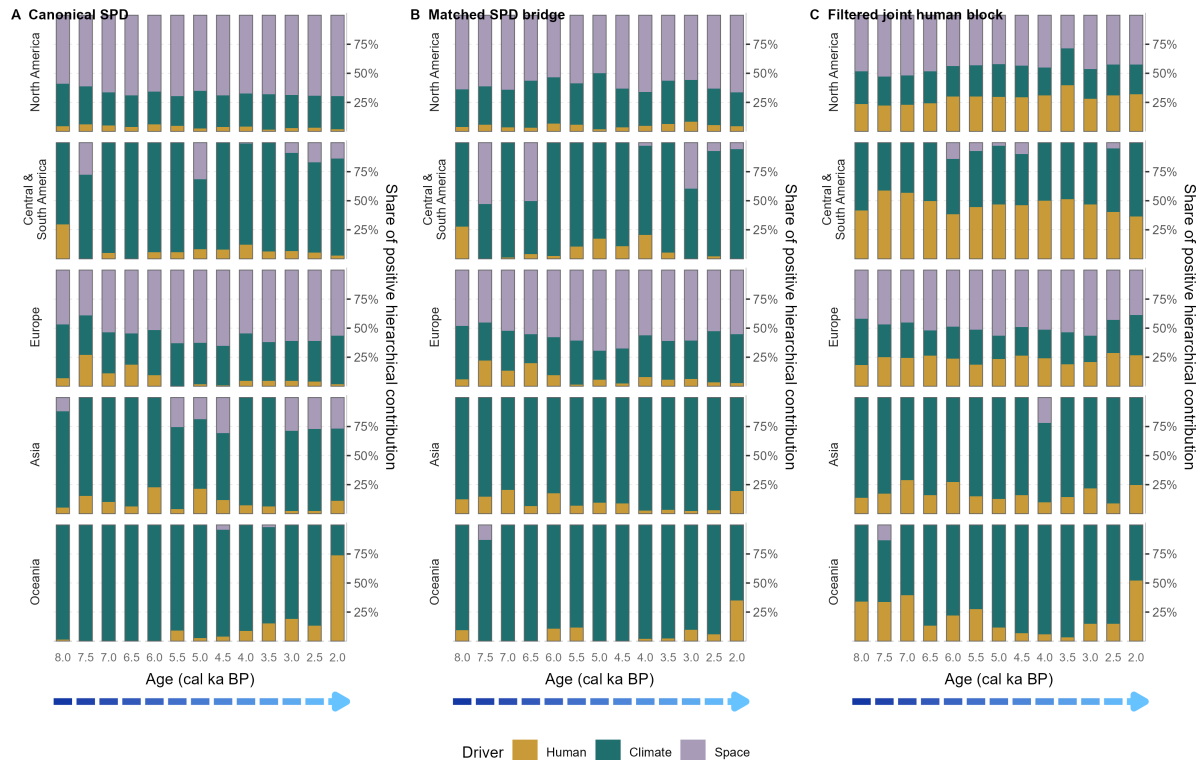


Figure 18: Figure 4 comparison on the exact common subset: canonical SPD, matched SPD bridge, and filtered joint human block. Every displayed stack sums to one after zero truncation.

16.5 Model fit, retained predictors, and the decision

Adding the filtered joint human block changes both allocation and model fit. Across the matched sequence comparison, median total adjusted (R^2) changes from 0.452 to 0.480. Pure human adjusted (R^2) generally increases, while the pure climate and time fractions often decrease, reflecting both the broader human block and changes in shared variation.

Predictor-retention frequencies vary among continental regions and regions. That variation is scientifically informative but complicates a simple statement that every local model represents exactly the same human construct.

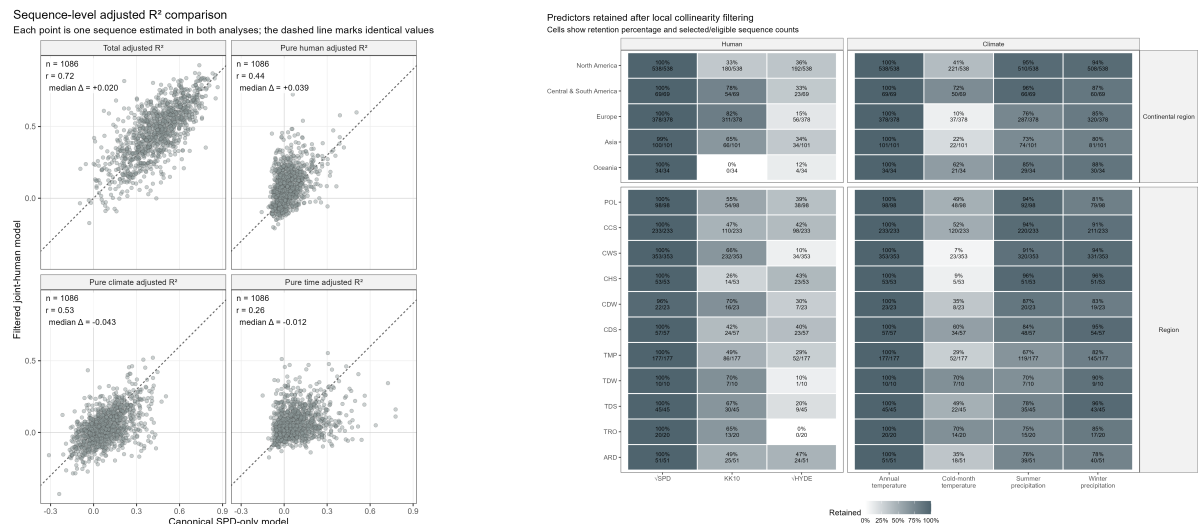


Figure 19: Sequence-level adjusted R-squared comparison (left) and local predictor retention by continental region and region (right).

Decision requested. The existing claim that climate generally outweighs human impact is not robust to a defensible expansion of the human predictor block. We need to agree what construct should represent human impact, how prominently both specifications should be shown, and whether H2 must be harmonised before the paper-wide narrative is revised.

16.5.1 Three defensible manuscript options

1. **Retain SPD as primary.** Keep canonical Figures 3–4, add the joint sensitivity, and substantially soften the climate-dominance headline. This fits current SPD-only H2 but retains a narrow construct.
2. **Replace primary H1 with the joint model.** Spatial H1 becomes human-dominant and temporal H1 remains mixed. A harmonised joint-proxy H2 refit is required.
3. **Treat both as co-primary.** Make sensitivity to the human construct the central finding and harmonise H2 across specifications.

16.5.2 Questions for the co-author meeting

1. Which construct best represents human impact: SPD alone or archaeological activity, land use, and modelled population together?
2. Is a scale- and specification-dependent conclusion preferable to one apparently definitive model?
3. Must H2 be harmonised before restructuring the title, abstract, and Discussion?
4. Which option is most transparent to reviewers and faithful to the evidence?

Requested meeting outcome: agree the human-impact construct, choose a provisional manuscript option, and decide whether to commission the H2 refit before changing the manuscript or reviewer response.

The memo intentionally makes no recommendation among the three options. Its purpose is to put the canonical revision and the new proxy sensitivity on the same evidential footing before the team decides.